Step 1: S_{Osm}

Click here to navigate

Step 2: U_{Osm}

Step 3: U_{Na}

Cases

Step 1: S_{Osm}

Step 2: U_{Osm}

Step 3: U_{Na}

Cases

Serum Osm

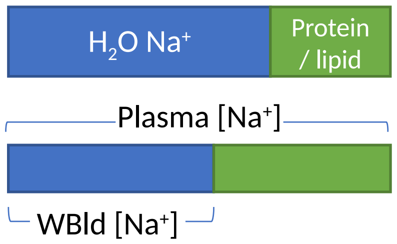
Is it real?

Na^+

>290 →

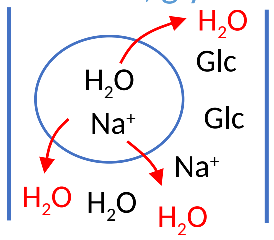
Artifact

↑ TGs or ↑ protein
(e.g. multiple myeloma)



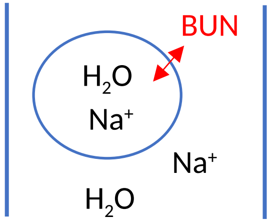
Active Osm

↑ glucose, mannitol, sorbitol, glycerin



Inactive Osm

BUN, ETOH



<290

Hypo-osmolar hyponatremia

Step 1: S_{Osm}

Step 2: U_{Osm}

Step 3: U_{Na}

Cases

Na^+

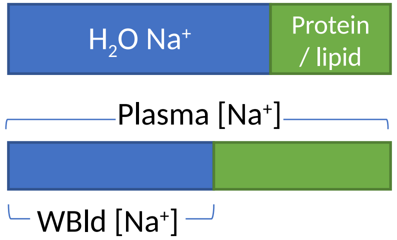
Serum Osm

Is it real?

>290

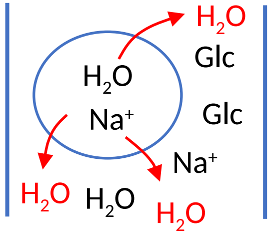
Artifact

↑ TGs or ↑ protein
(e.g. multiple myeloma)



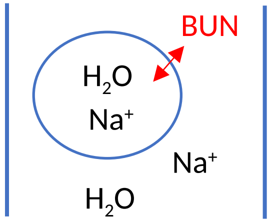
Active Osm

↑ glucose, mannitol, sorbitol, glycerin



Inactive Osm

BUN, ETOH



<290

Urine Osm

Is ADH on or off?

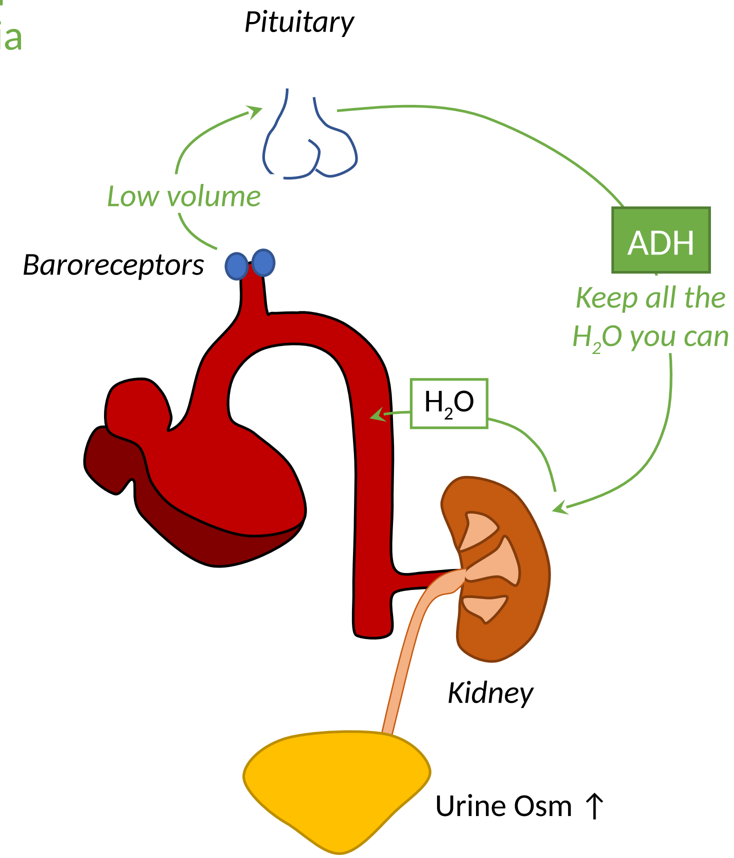
<100

OFF

Primary polydipsia
Beer potomania
Tea and toast

ON

>100



Step 1: S_{Osm}

Step 2: U_{Osm}

Step 3: U_{Na}

Cases

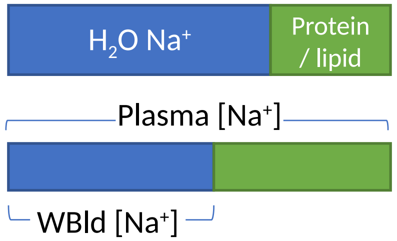
Na^+

Serum Osm
Is it real?

>290 →

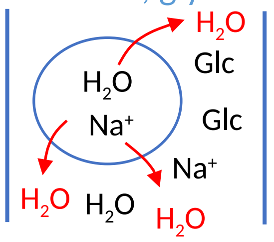
Artifact

↑ TGs or ↑ protein
(e.g. multiple myeloma)



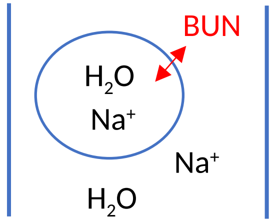
Active Osm

↑ glucose, mannitol, sorbitol, glycerin



Inactive Osm

BUN, ETOH



<290

Urine Osm
Is ADH on or off?

<100 →

Primary polydipsia
Beer potomania
Tea and toast

ON >100

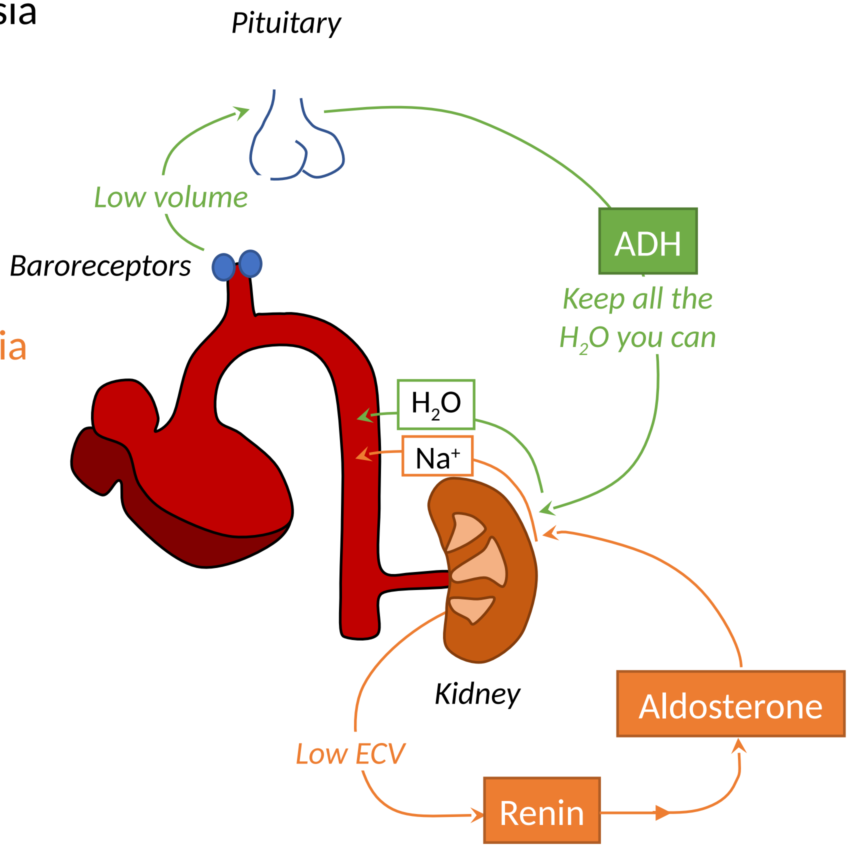
Urine Na^+
Are the kidneys seeing enough fluid?

<25 →

True hypovolemia
Heart failure
Cirrhosis

>40

Diuretics (e.g., loop, thiazide)
Adrenal Insufficiency
Hypothyroidism
SIADH



Step 1: S_{Osm}

Step 2: U_{Osm}

Step 3: U_{Na}

Cases

Case 1

40 yo F with nausea/vomiting and diarrhea. Serum Na is 125.

1. Serum osm 265
2. Urine osm 400
3. Urine Na < 10

Serum osm is low, or true hypoNa

Urine osm ↑ suggesting ↑ ADH state.

Urine Na is low suggesting low ECV state.

Given clinical history of nausea/vomiting and diarrhea, hyponatremia is likely from true hypovolemia.

Case 2

21 yo M with type I DM who presents with nausea/vomiting and abdominal pain. He presents with blood glucose of 500 and + ketones on UA. Serum Na is 125.

1. Serum osm 310
2. Urine osm 800
3. Urine Na 20

Hyperosm hypoNa. "something else in the blood." Glc is the active osm.
corrected Na = measured **Na** + [1.6 (**glucose** – 100) / 100]
The calculated Na is 131.

Urine osm ↑ suggesting ↑ ADH state.

Urine Na is borderline-low suggesting low ECV.

The patient has hyperosmolar hypoNa 2/2 to excess Glc. However, they also have true hypoNa (that is mild at 131). This hyponatremia is 2/2 to hypovolemia.

Case 3

68 yo M with a history of alcohol use disorder presents with alcohol withdrawal. Serum Na is 125.

1. Serum osm 270
2. Urine osm 60
3. Urine Na 65

Serum osm is low, or true hypoNa.

Urine osm is normal, suggesting ADH is off. This only occurs in poor solute intake or excess water intake.

Not needed in this case.

Given alcohol use history, this is likely beer potomania

Take-Home Points

1. Skip the volume-status guess — use the 3 labs: serum osm ("is it real?"), urine osm ("is ADH on?"), urine Na ("are the kidneys seeing enough fluid?").
2. True hyponatremia is HYPO-osmolar (SOsm < 290). Normal SOsm = pseudohyponatremia (lipids/paraprotein — send a whole-blood Na). High SOsm = something extra (glucose, mannitol): corrected Na = measured + 1.6 × (glucose − 100)/100.
3. Urine osm < 100 = ADH OFF — only primary polydipsia, beer potomania, or tea-and-toast (water in > solute in). Everything else is a high-ADH state.
4. ADH on + urine Na < 25 → the kidneys see low volume: true hypovolemia, or low effective circulating volume (HF, cirrhosis). ADH on + urine Na > 40 → kidneys see enough fluid: SIADH, hypothyroidism, adrenal insufficiency — or diuretics, which waste salt and break the rule.
5. This framework is diagnosis only — treatment (rate limits, hypertonic saline, overcorrection) is its own talk.

Attribution & Changelog

Original

"Evaluation of Hyponatremia — 3 Lab Approach to Diagnosis" — TeachIM.org (May 2020, updated Oct 2021). Authors: Brandon Fainstad MD, Yilin Zhang MD.

Corrections in this optimized copy (June 2026) — applied to BOTH boards

- Slide 4: "Aldolsterone" → "Aldosterone" (RAAS schematic label).
- Cases slide: "true hyponNa" → "true hypoNa".
- Appended slides only — all original slides, graphics, and click-to-reveal animations preserved on both boards.

Currency

3-lab diagnostic framework verified current June 2026 — the 2024 Austrian Society for Nephrology consensus uses the same serum osm / urine osm / urine Na triad. No changes to the standing teaching.

Companion files

Hyponatremia.pptx (learning board, corrected) · Hyponatremia-work-sheet.docx · Hyponatremia_Lesson_Package.html · Hyponatremia_OnePager.pdf · Hyponatremia_Anki_Cards.txt